

Left out of protection: social group, informality and earnings in Uttar Pradesh - evidence from PLFS 2025

Dopal Gupta
July 2026

Code and data documentation: [\[https://github.com/dopalgupta/UP-informality-plfs2025/blob/main/README.md\]](https://github.com/dopalgupta/UP-informality-plfs2025/blob/main/README.md)

Abstract

This research report examines the gaps across social groups in informal work and earnings in Uttar Pradesh (U.P.) using the Periodic Labour Force Survey (PLFS) (Jan–Dec 2025) microdata. There are 176,060 individuals, of whom roughly 64,000 are workers, representing a workforce of 68 million. There are three kinds of workers: the regular/salaried, the casual workers (wage earners) and the self-employed. A worker is classified as informal if they lack a written contract or social security (wage workers) or work in an unregistered enterprise (self-employed); estimates reproduce published PLFS benchmarks. Informality is near universal at 96%, but formal work is unequally distributed. Others workers are roughly three times as likely to hold protected jobs as SC/ST/OBC workers, with the gaps widest among the graduates. Formal workers earn nearly three times as much as informal workers, and SC, ST, and OBC workers earn 13–18% less than comparable Others workers; a penalty that, unlike the protection gap, survives all controls. Education explains who gets protection, but does not explain who gets paid less.

Introduction

In household surveys across low-income constituencies of Bihar, Delhi, and Madhya Pradesh, one pattern repeated itself with uncomfortable regularity: the families most in need of protection were the ones the systems of protection could not see. Below-poverty-line lists excluded eligible households; entitlements existed on paper but not in practice; and almost everyone worked long hours, in fields, shops, and construction sites, without a contract, a pension, or any cushion against a bad month. This note grew out of working with that survey data. It asks a simple question of India's largest state: who gets left out of good work and does education close the gap?

Uttar Pradesh is home to roughly 240 million people, if it were a country, it would be the world's fifth largest, and one of the largest concentrations of poor households in India. For most of its workers, employment offers no formal protection: no written contract, no social security, no registered employer. This matters for policy because employer-based social protection, the model on which formal safety nets rest, cannot reach workers the formal economy never admits. Understanding who is excluded from protected work, and why, is therefore a first-order question for social protection design in the state.

This note uses Periodic Labour Force Survey (PLFS) 2025 microdata to document social group gaps in informality and earnings among Uttar Pradesh's workers. It asks two questions. First, how is the rare good of formal, protected work distributed across social groups, and does education equalise access to it? Second, do earnings differ across social groups even among otherwise comparable workers? The answers, developed below, are: unequally, only partially, and yes.

Data and definitions

This note uses unit-level microdata from the Periodic Labour Force Survey (PLFS), calendar year 2025, for Uttar Pradesh. The working file covers 176,060 individuals in 36,774 households, built by merging the person- and household-level files on the survey's household identifiers. All estimates use the survey multipliers (rescaled per PLFS documentation) and represent roughly 188 million persons. Employment

follows the usual status (ps+ss) definition: a person is employed if their principal or subsidiary annual activity is work. As a validation check, this pipeline reproduces the published PLFS worker-population ratio for Uttar Pradesh (ages 15+) of 53.7% almost exactly, which anchors confidence in the weighting and definitions used throughout. The analysis universe is the roughly 64,000 workers classified by principal activity status, representing 68 million workers (Table 1).

Table 1. Sample and analysis universe, PLFS 2025, Uttar Pradesh

	Unweighted	Weighted (approx.)
Persons (all ages)	176,060	188 million
Households	36,774	39.9 million
Workers (principal status, classified)	63,844	68.1 million
of which informal	60,892	65.3 million
of which formal	2,952	2.8 million
Workers with positive earnings	47,096	49.1 million

Note: 52 self-employed workers with unclassifiable enterprise type are excluded from classification.

A worker is classified as informal following the logic of the ILO framework, with the test depending on employment type. Wage workers, regular/salaried and casual workers; are informal if they lack a written job contract or are not eligible for any employer-provided social security benefit; this deliberately strict criterion classifies a worker as formal only when both protections are present. Self-employed workers, who are not asked about contracts or benefits, are classified by their enterprise: those in proprietary or partnership enterprises are informal; those in government bodies, companies, cooperatives, or trusts are formal. Fifty-two self-employed workers with an unclassifiable enterprise type are excluded. Wage workers whose social security eligibility was recorded as “not known” are classified by contract status alone.

Monthly earnings are constructed per worker type: salaried workers' reported monthly earnings; self-employed workers' self-reported usual monthly earnings; and casual workers' weekly earnings from the daily-activity block, scaled by 4.33. The earnings analysis covers workers reporting positive earnings: about 47,000 persons. This excludes unpaid family helpers (about 9,600 workers, whose zero earnings are true by definition), casual workers with no work in the reference week (roughly one in five casual workers, itself a measure of precarity), and a small share of non-reporters. Social groups follow the PLFS classification: Scheduled Tribes (ST), Scheduled Castes (SC), Other Backward Classes (OBC), and Others. Education is grouped into five levels, from no formal schooling to graduate and above. Nine transgender respondents are excluded from sex-disaggregated analysis owing to sample size.

Findings

Formal work in Uttar Pradesh is rare and rarest for SC, ST, and OBC workers. 96% of Uttar Pradesh's workers are informal. Speaking of the informality across social groups with weighted means, we find that workers of the Others group have an informality rate of 90.9%, OBC have an informality rate of 97.1%, SC have an informality rate of 96.5%, and ST have an informality rate of 96.2% (Table 2). Flipping it to formality: Others 9.1% vs OBC 2.9%, SC 3.5%, ST 3.8%. We find that the Others workers are 3 times more likely to find a formal job as compared to OBC, 2.6 times compared to SC, and 2.4 times compared to ST workers.

Table 2. Informality among workers, by social group (weighted, %)

Social group	Informal (%)	Formal (%)
ST	96.2	3.8

SC	96.5	3.5
OBC	97.1	2.9
Others	90.9	9.1
All workers	95.9	4.1

Source: author's calculations from PLFS 2025 microdata.

Education raises the chance of formal work for everyone — but unequally. Below secondary education, informality exceeds 98% for every group, and gaps between groups are small and unordered. Among graduates, informality is 79.4% for Others workers against 86.9 -- 90.9% for OBC, SC, and ST workers (Figure 1, Table 3). The gap between groups is thus negligible at low education levels and widens to 8–11 percentage points among graduates: the higher the education level, the more unequal the access to protected work. Estimates for ST workers rest on a small sample and should be read with caution, in some education cells, no formal ST worker was sampled at all.

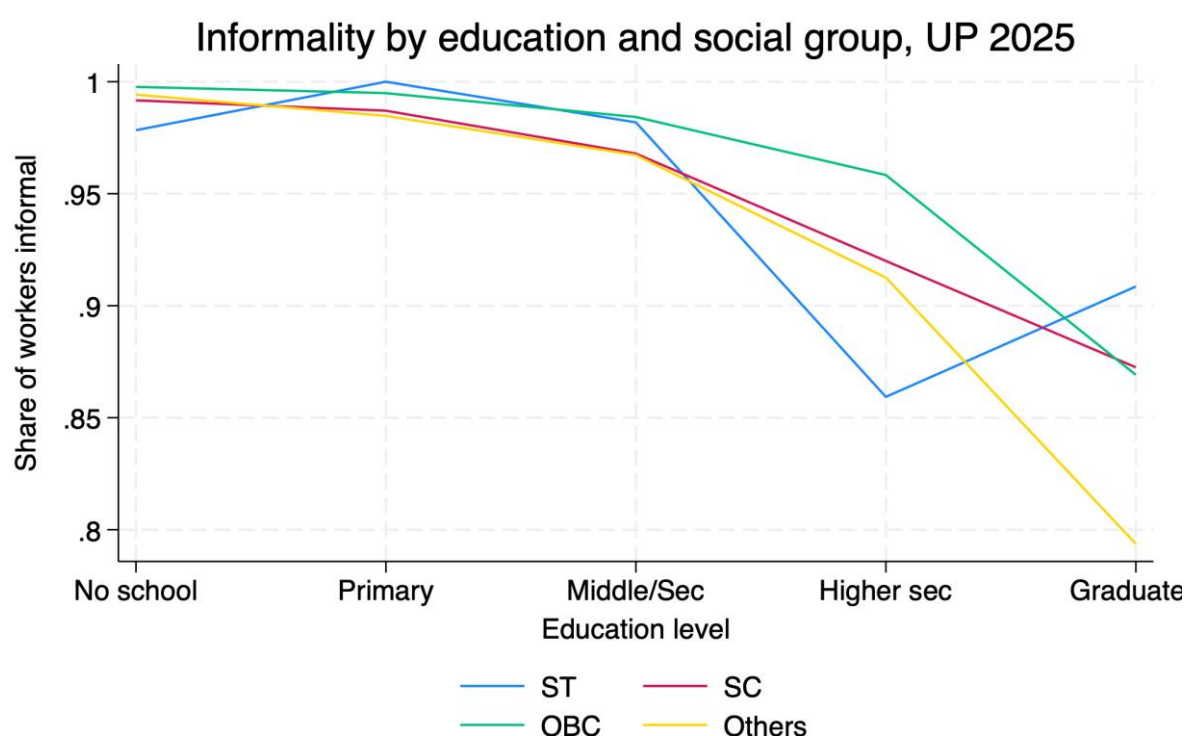


Figure 1. Share of workers in informal employment, by education level and social group, UP 2025. Source: author's calculations from PLFS 2025. ST series based on small samples; see Caveats.

Table 3. Informality by social group and education level (weighted, %)

Education level	ST	SC	OBC	Others
No formal schooling	97.8	99.2	99.8	99.4
Primary or below	100.0	98.7	99.5	98.5
Middle/Secondary	98.2	96.8	98.4	96.7
Higher secondary	85.9	92.0	95.8	91.3
Graduate and above	90.9	87.3	86.9	79.4

Note: ST estimates rest on small samples; at primary level, no formal ST worker was sampled.

The earnings hierarchy mirrors the protection hierarchy. The Others group earn ₹18,832/month on average compared to the OBC group at ₹12,498/month, ST workers at ₹10,852/month and SC workers

at ₹10,718/month (Table 4). The earnings disparity across social groups is greatest among salaried workers. In this category, the Others group earns substantially more than ST, SC, and OBC workers, with the largest gap observed between Others and ST workers (approximately ₹10,573 per month). By contrast, earnings differences are much smaller among casual workers (₹9,797–10,629 across all four groups). Where pay is set by the daily labour market, everyone earns equally little; where jobs are good, access to the best-paying ones is unequal.

Table 4. Mean monthly earnings (₹), workers with positive earnings - Panel A: by social group and worker type

Social group	All workers	Salaried	Self-employed	Casual
ST	10,852	14,126	10,021	10,367
SC	10,718	15,401	9,638	9,797
OBC	12,498	16,988	11,785	10,253
Others	18,832	24,699	16,778	10,629

Source: author's calculations from PLFS 2025 microdata.

Protection and pay travel together. Formal workers earn ₹34,646 per month on average: nearly three times the ₹11,850 earned by informal workers. These are means, and earnings are right-skewed: a small number of high earners pulls the formal average upward, so the ratio should be read as indicative; the regression analysis below uses log earnings partly for this reason. Even so, the direction and rough size of the gap are unambiguous: informality in Uttar Pradesh is not a paperwork distinction but a threefold difference in living standards.

Table 4 - Panel B: by informality status

Status	Mean monthly earnings (₹)
Formal	34,646
Informal	11,850
Ratio	2.9

Source: author's calculations from PLFS 2025 microdata.

Regression evidence

To test whether the gaps described above survive fair comparison, three regressions were estimated. The first is a linear probability model of informality on social groups alone, measuring raw gaps. The second adds controls for education, age, sex, and urban location, asking how much of each gap remains between otherwise similar workers. The third estimates the same specification using log-monthly earnings as the outcome for workers reporting positive earnings. All models use survey weights with robust standard errors; the Others group is the comparison category throughout (Table 5).

For protection, raw social-group gaps are largely attributable to differences in observed characteristics. Without controls, ST, SC, and OBC workers are 5.3, 5.6, and 6.2 percentage points more likely than Others workers to be in informal work (all $p < 0.01$). With controls, the ST gap disappears entirely (-0.03 points, $p = 0.99$) and the SC gap falls to 0.9 points and loses statistical significance ($p = 0.16$). The OBC gap, however, survives at 2.7 percentage points ($p < 0.001$). For protection, raw social-group gaps are largely attributable to differences in observed characteristics. Without controls, ST, SC, and OBC workers are 5.3, 5.6, and 6.2 percentage points more likely than Others workers to be in informal work (all $p < 0.01$). With controls, the ST gap disappears entirely (-0.03 points, $p = 0.99$) and the SC gap falls to 0.9 points and loses statistical significance ($p = 0.16$). The OBC gap, however, survives at 2.7 percentage points ($p < 0.001$), unexplained by education, age, sex, or location. These average results conceal the pattern shown earlier: gaps are negligible at low education levels and widen sharply among

the educated, a heterogeneity the additive model averages over. Among the controls, urban residence (-7.3 points) and graduate education (-13.8 points relative to no schooling) are strongly associated with formal work. unexplained by education, age, sex, or location. These average results conceal the pattern shown earlier: gaps are negligible at low education levels and widen sharply among the educated, a heterogeneity the additive model averages over. Among the controls, urban residence (-7.3 points) and graduate education (-13.8 points relative to no schooling) are strongly associated with formal work.

For earnings, the penalties survive for every group. Controlling for education, age, sex, and location, SC workers earn about 18% less than comparable Others workers, OBC workers about 15% less, and ST workers about 13% less (all $p \leq 0.001$). The contrast with the protection results is the central finding of this note: observed characteristics largely explain who obtains protected work, but they do not explain who is paid less. The largest coefficient in the model belongs to sex For earnings, the penalties survive for every group. Controlling for education, age, sex, and location, SC workers earn about 18% less than comparable Others workers, OBC workers about 15% less, and ST workers about 13% less (all $p \leq 0.001$). The contrast with the protection results is the central finding of this note: observed characteristics largely explain who obtains protected work, but they do not explain who is paid less. The largest coefficient in the model belongs to sex - women earn roughly 54% less than comparable men, a gap outside this note's scope but too large to leave unremarked.

Two limitations apply. The linear probability model can produce fitted probabilities outside the unit interval and is used here for interpretability. The earnings regression covers only workers reporting positive earnings, excluding unpaid family helpers by construction.

Table 5. Regression results

	(1) Informal	(2) Informal	(3) Log earnings
ST	0.053** (0.018)	-0.000-0.000 (0.018)	-0.137**-0.137** (0.042)
SC	0.056*** (0.006)	0.009 (0.006)	-0.201***-0.201*** (0.016)
OBC	0.062*** (0.005)	0.027*** (0.005)	-0.157***-0.157*** (0.013)
Female		-0.004-0.004 (0.002)	-0.772***-0.772*** (0.016)
Age		-0.001***-0.001*** (0.000)	0.001 (0.000)
Urban		-0.073***-0.073*** (0.006)	0.359*** (0.010)
Primary or below		-0.004-0.004 (0.002)	0.094*** (0.019)
Middle/Secondary		-0.020***-0.020*** (0.003)	0.174*** (0.015)
Higher secondary		-0.056***-0.056*** (0.005)	0.278*** (0.018)
Graduate and above		-0.138***-0.138*** (0.005)	0.604*** (0.019)
Constant	0.909*** (0.005)	1.015*** (0.008)	9.170*** (0.028)
N	63,844	63,842	47,095
R ²	0.013	0.092	0.329

Notes: Robust standard errors in parentheses. Survey (probability) weights used. Comparison group: Others (social group); no formal schooling (education). Models (1)–(2): linear probability models of informality. Model (3): log monthly earnings, workers with positive earnings. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Caveats

Five limitations should be kept in view. First, the descriptive estimates use importance weights, whose standard errors do not account for PLFS's clustered survey design; the reported means are correct, but their precision is overstated, and the regression estimates with robust standard errors should be preferred for inference. Second, estimates for ST workers rest on a small sample (about 1,700 persons in the UP file) and should be read with caution, in some education cells, no formal ST worker was sampled at all. Third, self-employed earnings are self-reported perceived monthly earnings, and are measured less precisely than salaries; casual workers' monthly earnings are constructed from one reference week (weekly earnings $\times$ 4.33), which captures work irregularity but assumes the surveyed week is representative. Fourth, for workers whose social security eligibility was recorded as “not known” (about 3,000 wage workers), classification rests on contract status alone, which may modestly understate informality. Fifth, this is a descriptive note based on a single cross-section: the gaps documented here are associations, not causal estimates, and characteristics not observed in the survey - wealth, networks, land, discrimination, may lie behind the unexplained penalties.

Conclusion

Informality in Uttar Pradesh is near-universal, but its rare exception, protected, formal work is distributed along social group lines, and increasingly so at higher education levels. Education emerges as the strongest observed route into protected work, yet it does not equalise access among the educated, and it does not close earnings gaps at all: SC, ST, and OBC workers earn 13–18% less than comparable Others workers even with identical observed characteristics. For social protection policy, the implication is twofold. Expanding education will raise formality but will not, by itself, deliver equal access to protection or pay. And because 96% of the state's workers remain outside employer-based protection entirely, the binding question for Uttar Pradesh is less how workers move into formal jobs than how protection reaches the informal economy, where nearly everyone works.
